Mid-Project Report

Predicting Student Academic Success Using Machine Learning

COMP 1100 | Konley Stoup and Camden Ball | October 27, 2026

1. Project Overview

Our project uses machine learning to predict whether a student will pass or fail a course. One challenge in education is identifying struggling students early enough for teachers to provide additional support. If teachers can recognize students who may be at risk of failing, they can offer tutoring, academic resources, or other assistance before the end of the course.

Our goal is to determine whether machine learning can accurately predict student pass/fail outcomes, and whether including grades from earlier in the course improves those predictions. We compare two machine-learning algorithms using two different sets of student information to see which approach performs best.

2. Dataset and AI Technologies

We are using the Student Performance dataset from the UCI Machine Learning Repository, which contains information about 395 mathematics students from two secondary schools in Portugal, including study time, previous class failures, absences, family background, and grades.

We used Microsoft Excel to prepare the data and created a new column called **PassFail**. Students with a final grade (G3) of 10 or higher on the 0–20 scale were labeled Pass; students below 10 were labeled Fail. This produced 265 passing and 130 failing students. We then created two versions of the dataset:

- **Dataset A – Without Grades:** 30 student characteristics plus PassFail, excluding all grading-period grades (G1, G2, and G3).
- **Dataset B – With G1:** the same characteristics plus the first-period grade (G1), excluding G2 and G3.

We used **WEKA 3.8.6** to train and evaluate two algorithms: the J48 Decision Tree and Logistic Regression. A decision tree makes predictions by following a series of if/then rules learned from the data, while logistic regression estimates the probability that a student belongs to each class. We tested both algorithms on both datasets, for four experiments in total. Each used 10-fold stratified cross-validation, which splits the data into ten parts and repeatedly trains on nine parts while testing on the remaining one, so every student is evaluated by a model that never saw them during training.

3. Work Completed and Preliminary Results

We have completed data preparation and trained and evaluated all four configurations in WEKA. We saved the full WEKA output for each run and captured screenshots documenting the experiments.

To judge whether the models are useful, we compare them to a simple **baseline** that predicts “Pass” for every student. Because 265 of 395 students passed, this baseline is already 67.1% accurate while identifying none of the failing students, so a useful model must beat it, especially on catching students who fail.

Model	Accuracy	Fail Recall	Fail Precision	ROC Area
Baseline (always Pass)	67.1%	0.000 (0/130)	–	0.500
J48 – Without Grades	66.3%	0.369 (48/130)	0.485	0.572
Logistic – Without Grades	68.6%	0.369 (48/130)	0.533	0.672
J48 – With G1	84.3%	0.777 (101/130)	0.754	0.820
Logistic – With G1	84.3%	0.769 (100/130)	0.758	0.921

Table 1. Pass/fail prediction performance using 10-fold cross-validation. Fail recall is the share of failing students the model correctly identified; fail precision is the share of students flagged as Fail who actually failed.

Without earlier grades, J48 reached 66.3% accuracy and Logistic Regression 68.6%, essentially the same as the baseline. Both models correctly identified only 48 of the 130 students who actually failed, missing 82. Student characteristics and habits alone therefore provided limited predictive power. The strongest signal the models found was the number of past class failures: it was the first split in the J48 tree, and in Logistic Regression each previous failure multiplied the odds of failing by about 2.4.

When we added the first-period grade (G1), both algorithms reached 84.3% accuracy, a large improvement. J48 correctly identified 101 of the 130 failing students and Logistic Regression identified 100, roughly double the models without grades. For example, Logistic Regression with G1 correctly flagged 100 failing students, missed 30, and incorrectly flagged 32 students who actually passed. Logistic Regression with G1 also achieved the highest ROC area (0.921), meaning it was the best of our models at ranking students from highest to lowest risk.

The J48 tree with G1 shows how important the early grade is. Its first split is on G1: in the model built from the full dataset, students with G1 above 10 were predicted to pass (202 students, with 5 errors), while students with G1 of 7 or below who received no extra school support were predicted to fail (54 students, with 2 errors). These findings support our expectation that first-period grades provide valuable information for predicting final outcomes.

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=== Stratified cross-validation ===
=== Summary ===

Correctly Classified Instances         262               66.3291 %
Incorrectly Classified Instances       133               33.6709 %
Kappa statistic                          0.1882
Mean absolute error                      0.3865
Root mean squared error                  0.5351
Relative absolute error                 87.4549 %
Root relative squared error            113.8802 %
Total Number of Instances              395

=== Detailed Accuracy By Class ===

               TP Rate  FP Rate  Precision  Recall   F-Measure  MCC      ROC Area  PRC Area  Class
               0.369    0.192    0.485      0.369    0.419      0.192    0.572     0.399     Fail
               0.808    0.631    0.723      0.808    0.763      0.192    0.572     0.693     Pass
Weighted Avg.  0.663    0.487    0.645      0.663    0.650      0.192    0.572     0.596

=== Confusion Matrix ===

   a   b   <-- classified as
  48  82 |   a = Fail
  51 214 |   b = Pass
```

Figure 1. WEKA J48 results without grades (Dataset A): 66.3% accuracy, 48 of 130 failing students identified.

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=== Stratified cross-validation ===
=== Summary ===

Correctly Classified Instances         333               84.3038 %
Incorrectly Classified Instances        62               15.6962 %
Kappa statistic                          0.6459
Mean absolute error                      0.1798
Root mean squared error                  0.3353
Relative absolute error                 40.6935 %
Root relative squared error             71.3664 %
Total Number of Instances              395

=== Detailed Accuracy By Class ===

               TP Rate  FP Rate  Precision  Recall   F-Measure  MCC      ROC Area  PRC Area  Class
               0.769    0.121    0.758      0.769    0.763      0.646    0.921     0.837     Fail
               0.879    0.231    0.886      0.879    0.883      0.646    0.921     0.963     Pass
Weighted Avg.  0.843    0.195    0.844      0.843    0.843      0.646    0.921     0.921

=== Confusion Matrix ===

   a   b   <-- classified as
 100  30 |   a = Fail
  32 233 |   b = Pass
```

Figure 2. WEKA Logistic Regression results with G1 (Dataset B): 84.3% accuracy, 100 of 130 failing students identified, ROC area 0.921.

4. What Is Working, Challenges, and Limitations

Working: the full pipeline, from raw data to Excel preparation to WEKA training and cross-validated evaluation, runs correctly for all four configurations, and the comparison clearly answers our research question.

Challenges: preparing the data in Excel took several attempts. Initially, PassFail was a formula referencing G3, so the labels disappeared when we deleted G3; we fixed this by converting the formulas to fixed values first. Our first CSV export also contained blank columns where grades had been cleared rather than deleted, which we removed before loading the data into WEKA. Finally, we checked that the two datasets lined up across all 395 students, with no mismatches in their PassFail labels.

Not yet working well: the models still make important mistakes. Even with G1, about 30 of the 130 failing students are predicted to pass. Since identifying students who need support is the main purpose of the project, these missed students are the most serious type of error.

Limitations: the data represents Portuguese secondary-school students, not American college students, so our results cannot be assumed to apply at Auburn. Some attributes, such as total absences, may not be known by the end of the first grading period, which affects how early a real prediction could be made. Some inputs are also personal or sensitive (for example sex, family situation, and alcohol use), raising fairness and privacy concerns. Any real-world use would require teachers to treat predictions as guidance, not as decisions.

5. Remaining Work and Next Steps

Before the final submission we will create Excel charts comparing accuracy and fail recall across the models, examine incorrectly classified students in more detail, and further investigate which characteristics matter most and why (being careful to separate correlation from causation). We will then write the final report covering our methods, results, limitations, responsible-use considerations, and lessons learned, and create a one-slide poster summarizing the project.

6. Conclusion

Our project has moved from an idea to a working machine-learning prototype: we prepared two datasets, trained four model configurations, and evaluated them in WEKA. Models without earlier grades performed close to the always-Pass baseline, while both models using G1 reached 84.3% accuracy and identified roughly twice as many failing students. Machine learning may help identify students who could benefit from extra support, especially once early grades are available, but its predictions are imperfect and require careful human interpretation.

References

Cortez, P., & Silva, A. (2008). Using data mining to predict secondary school student performance. In *Proceedings of the 5th Future Business Technology Conference (FUBUTEC 2008)* (pp. 5–12). EUROSIS. Dataset: UCI Machine Learning Repository, https://archive.ics.uci.edu/dataset/320/student+performance

Frank, E., Hall, M. A., & Witten, I. H. (2016). *The WEKA Workbench*. Online appendix for *Data Mining: Practical Machine Learning Tools and Techniques* (4th ed.). Morgan Kaufmann.